

Supplementary Material 1: Practical Implementation Guidelines

Description: *The proposed consistent SLAM approach is designed for broad applicability across arboreal agricultural environments, but its performance is influenced by several orchard-specific characteristics. This supplementary material discusses the method’s generalizability and scope, describes how environmental factors affect localization and mapping, and provides practical recommendations for configuring and tuning the algorithm’s parameters when deploying on a new orchard type or robotic platform.*

1. Generalizability and scope

The core design principle of the proposed approach, projecting 3D lidar scans onto a 2D ground plane and matching via the modified Hausdorff distance, is applicable to any environment where vertically elongated, spatially persistent structures (e.g., tree trunks, vineyard posts, plantation stakes) are present at approximately known heights above the ground. This encompasses a wide range of arboreal and semi-structured agricultural environments, including citrus and stone-fruit orchards, nut plantations, vineyards with trellis systems, and forest plantations with defined row structure.

A key strength of the approach is its independence from IMU, GNSS, and wheel odometry. This makes it directly applicable to handheld mapping devices, quadruped robots, and humanoid platforms, where discontinuous leg-ground contact forces corrupt inertial measurements and render IMU-based odometry unreliable. Because the scan matching is solved in 2D and initialized from the EKF-predicted pose $\hat{\mathbf{x}}_{t|t-1}$, the method is also computationally efficient and can run in real time on modest embedded hardware. Importantly, the method is *not sensor-specific*: it interfaces with the lidar hardware only through two scalar quantities, the range standard deviation σ_s , which characterizes measurement noise (Eq. 4), and the vertical angular resolution, which determines how many elevation rings intersect the trunk at a given operating range (see Step 2 of Section 3 of this document), both of which are available directly from the manufacturer’s datasheet. No modification to the scan matching (Eqs. (11)–(12)) or EKF equations is required when switching lidar models; only σ_s , $\bar{z}$ (and hence $\lambda = \ln(2)/\bar{z}$), Δm , and D_{max} need to be set accordingly, following the calibration procedure of Section 3 of this document.

The approach naturally handles low-lying vegetation. The probabilistic height filter defined in Eqs. (5) and (6) assigns higher likelihood to measurements consistent with tree trunks near the ground, and lower likelihood to points higher in the canopy. Crucially, a low-lying return (e.g. from grass or ground cover) will only be retained if it is also consistent with an occupied cell in the existing map, that is, if

$$p_d(D(T_s^w(x_t^s, y_t^s; \mathbf{x}_t), \mathcal{M}_{t-1})) \geq \beta_d.$$

Isolated grass returns that do not correspond to persistent map structures are therefore naturally down-weighted during scan matching via the inlier rejection mechanism governed by the parameter K in Eqs. (11) and (12). However, if tall weed mounds (e.g., exceeding approximately 0.5 m) are spatially consistent across consecutive scans, they may be incorporated into the map as persistent structures. This does not necessarily degrade localization, provided that the weed patterns are stable between scans, they constitute valid landmarks, but it does reduce the interpretability of the map as a representation of permanent orchard infrastructure; see for example Fig. 1, which shows in the upper-right image a large mound of chipped wood in the path of the robot. In environments with dense tall ground vegetation, it is therefore recommended to set the canopy height parameter $\bar{z}$ conservatively (i.e., equal to or slightly below the minimum expected trunk height) so that the height filter p_h in (5) excludes a larger fraction of low-lying returns.

The method is less suitable for environments where tree trunks are absent or indistinct as vertical structures. For example, in very young orchards with very thin trunks (diameter less than 5 cm), or in crops trained as horizontal espaliers (flat against a wall or wire), the 2D projection of lidar scans may not yield sufficiently distinct or stable point distributions for reliable scan matching. Similarly, environments with very sparse tree spacing (e.g., inter-row spacing exceeding the lidar’s effective matching range) may result in insufficient overlap between the current scan $\bar{\mathcal{Z}}_t^w(\mathbf{x}_t)$ and the map $\mathcal{M}_{t-1}$, leading to ambiguous pose estimates.

2. Influence of environmental characteristics on performance

The main environmental variables influencing localization and mapping performance are: (i) tree height and trunk diameter; (ii) canopy density; (iii) ground flatness and terrain roughness; and (iv) row structure and inter-row spacing. These are discussed in turn.

Tree height and trunk diameter: The height filter parameter $\lambda = \ln(2)/\bar{z}$ in Eq. (5) controls the vertical extent of the lidar slice used for 2D projection. Taller trees with well-defined trunks (e.g., mature citrus, plum, or almond trees) provide a longer vertical trunk section, increasing the number of consistent projected points per scan and thereby improving matching robustness. Short-canopy crops (e.g., young vines or dwarf rootstock apple trees) reduce the effective number of contributing scan rings, potentially degrading localization. Trunk diameter also matters: thicker trunks produce broader, more stable point clusters in the projected set $\bar{\mathcal{Z}}_t^w(\mathbf{x}_t)$, while very thin trunks (saplings) may be confused with measurement noise and rejected by the inlier selection governed by K in (11).

Canopy density: Dense canopies increase the proportion of lidar returns originating from foliage rather than trunks. These points are located at heights above $\bar{z}$ and are therefore attenuated by p_h , Eq. (5), but they are not fully eliminated since the filter is probabilistic. In very dense canopies, the proportion of outlier foliage returns increases, and the inlier ratio α (defined in Section 2.1 of the paper as $\alpha = K/|\bar{\mathcal{Z}}_t^w(\hat{\mathbf{x}})| \in [0.7, 0.8]$) must be set conservatively (toward the lower end of that range) to prevent foliage noise from corrupting the scan-to-map matching in (11). Seasonal variation in canopy density (e.g., deciduous trees in winter versus summer) can change the density of useful trunk returns and may require seasonal adjustment of $\bar{z}$ and α .

Ground flatness and terrain roughness: The proposed method operates on a 2D ground-plane projection and uses a differential-drive constant-velocity motion model in the EKF prediction step [1]. On flat terrain, this model provides an accurate initial guess for the scan matching optimizer, which typically needs to search within 0.5 *m* and 0.2 *rad* of the true solution. On rough terrain (e.g., furrows, ruts, or root mounds), the effective heading of the lidar sensor may deviate from the robot’s nominal heading due to platform roll and pitch, introducing apparent rotational errors in the projected scan $\bar{\mathcal{Z}}_t^w(\mathbf{x}_t)$. These errors manifest as misalignment between consecutive projected scans, degrading the Hausdorff distance minimization in (11). In practice, this impact is mitigated by the robustness of the modified Hausdorff distance to outliers and by the EKF correction step [1], but for highly uneven terrain an additional pitch/roll compensation step using, for example, a tilt sensor or proprioceptive leg-state estimates from a legged robot, would further reduce apparent scan misalignment.

Row structure and inter-row spacing: The proposed approach performs best in environments with well-defined, regularly spaced rows of trees, as in citrus orchards or forest plantations. The periodic structure of tree rows provides strong, unambiguous scan matching constraints when the robot traverses along a row. At row ends during turns, however, the lidar field of view may transition between adjacent rows of trees, temporarily reducing the number of consistent matched points in $\bar{\mathcal{Z}}_t^w(\hat{\mathbf{x}}; K)$. This effect contributes to the slightly higher heading estimation errors observed at turns in the Pullally dataset (Table 11). Orchards with very narrow inter-row spacing (less than the footprint of the lidar’s horizontal field at the trunk height) may cause range ambiguity between adjacent rows; in such cases, increasing Δm and decreasing D_{max} in (12) can help maintain matching specificity.

3. Step-by-step parameter calibration procedure

The following procedure is recommended for deploying and calibrating the proposed approach in a new orchard environment. It requires only basic field measurements obtainable with a tape measure or a single lidar scan, and can typically be completed before deployment.

1. **Characterize the orchard geometry.** Collect the following measurements from the target orchard: (a) average tree trunk height to the first main branch or to canopy closure, denoted $\bar{z}$ [m]; (b) average trunk diameter at sensor height [m]; (c) inter-tree spacing along the row and inter-row spacing [m]; and (d) approximate terrain roughness class: *flat* (< 5 *cm* relief over 1 *m*), *moderate* (5–15 *cm*), or *rough* (> 15 *cm*).
2. **Set the height filter parameter λ .** Set the canopy height parameter to $\bar{z}$ as measured in Step 1 and compute $\lambda = \ln(2)/\bar{z}$ in (5). In environments with significant low-lying vegetation (weeds or cover crops exceeding 30 *cm*), reduce $\bar{z}$ by 10–20% to bias p_h toward lower trunk returns and minimize inclusion of weed structures.

Verify that the resulting filter selects at least two lidar elevation rings intersecting the trunk at the expected operating range r . For the Velodyne VLP-16 with a 2° vertical angular resolution, at $r = 3\text{ m}$ and $\bar{z} = 1.5\text{ m}$, approximately 14 elevation rings intersect the trunk, which is more than sufficient. If fewer than three rings contribute, increase $\bar{z}$ slightly or reduce the minimum considered range.

3. **Set the map resolution Δm and D_{max} .** Set the grid map resolution Δm to be at least as large as the lidar's range accuracy (e.g., $\Delta m = 0.05\text{ m}$ for the Velodyne VLP-16 with $\pm 3\text{ cm}$ accuracy). Set $D_{max} \in [2\Delta m, 4\Delta m]$ (i.e., $0.10\text{--}0.20\text{ m}$ for the default resolution), as discussed in Section 2.1 of the paper. In environments with thin trunks or short inter-tree spacing, reduce D_{max} toward $2\Delta m$ to increase matching specificity. In open environments with sparse trees or rough terrain, increase D_{max} toward $4\Delta m$ to accommodate larger initial misalignments.
4. **Set the lidar range variance σ_s^2 .** Set σ_s to the manufacturer-specified range accuracy (e.g., $\sigma_s = 0.03\text{ m}$ for the Velodyne VLP-16). The threshold $\beta_d = e^{-1/2} / \sqrt{2\pi\sigma_s^2} \approx 1/(4\sigma_s)$ in (6) then follows automatically, selecting measurements within $\pm 1\sigma$ of the expected range.
5. **Set the inlier ratio α .** Begin with $\alpha = 0.75$ (the midpoint of the recommended range $[0.7, 0.8]$). In dense-canopy environments with high foliage noise, reduce toward $\alpha = 0.70$; in open or winter-canopy environments with sparse returns, increase toward $\alpha = 0.80$. The sensitivity of localization to α can be assessed by running the algorithm on a short loop-closure trajectory (e.g., a rectangle around a single tree row) and evaluating the loop-closure error under different values of α .
6. **Set the EKF process noise covariance $\mathbf{Q}_t$.** Set $\mathbf{Q}_t$ based on the platform dynamics [1]. For wheeled robots on flat terrain, translational noise of $0.02\text{--}0.05\text{ m}$ per step and rotational noise of $0.01\text{--}0.02\text{ rad}$ per step are appropriate. For legged robots on rough terrain, increase translational noise to $0.05\text{--}0.15\text{ m}$ and rotational noise to $0.05\text{--}0.10\text{ rad}$ per step, reflecting the larger pose uncertainty between scans due to impulsive leg-ground contact forces. These values bound the expected search range for the scan matching optimizer and should be set conservatively (i.e., slightly larger than the expected worst-case step error) to ensure that the initialization from $\hat{\mathbf{x}}_{t-1}$ is always within the basin of convergence of the correct solution.
7. **Validate on a short trial trajectory.** Before executing the full mission, run a short validation trajectory (one or two orchard rows) with the GNSS active if available. Compute the mean position estimation error $\bar{e}_p$ (14) and inspect the reconstructed 2D map $\mathcal{M}_t$ visually: tree trunk clusters should appear as compact, well-separated blobs consistent with the known orchard layout. If the map shows blurred or doubled trunk clusters, reduce α or increase σ_s . If trunk clusters appear fragmented or absent, increase $\bar{z}$ or D_{max} .

4. Summary of key tunable parameters

Table 1 summarizes the principal tunable parameters, their recommended defaults, and guidance on when to increase or decrease each parameter to adapt the method to a new deployment environment and robotic platform.

References

- [1] Peralta-Cabezas, J.L., Torres-Torriti, M., Guarini-Hermann, M., 2008. A comparison of Bayesian prediction techniques for mobile robot trajectory tracking. *Robotica* 26, 571–585. doi:10.1017/S0263574708004153.

Table 1: Summary of tunable parameters of the proposed SLAM approach, with recommended defaults and adjustment guidance for different orchard environments and robotic platforms.

Parameter	Definition	Default	Increase if...	Decrease if...
$\bar{z}$ [m]	Avg. trunk / canopy height	Measured in field	Open canopy; tall, mature trunks	Dense low vegetation; tall weeds present
λ	Height filter rate = $\ln(2)/\bar{z}$, Eq. (5)	Derived from $\bar{z}$	Derived automatically (see Step 2)	
$\alpha \in [0.7, 0.8]$	Inlier ratio $K/ \overline{\mathcal{Z}}_t^w $	0.75	Sparse canopy; winter; few foliage returns	Dense canopy; summer; many foliage returns
Δm [m]	Grid map resolution	0.05	Wide trunks; coarse survey needed	Thin trunks; narrow inter-tree spacing
D_{max} [m]	Max point-to-map distance, Eq. (12)	$3 \Delta m$	Rough terrain; large step-to-step pose errors	Sparse rows; risk of false cross-row matches
σ_s [m]	Lidar range std. dev., Eq. (4)	Sensor spec (e.g., 0.03)	Noisy sensor; long-range operation	Short-range, high-accuracy sensor
$\mathbf{Q}_t$	EKF process noise covariance	Platform-specific	Legged robot; rough terrain; fast gait	Wheeled robot; smooth terrain; slow speed